

Smoothness Ranks Skill, Not Success: An Audit of Published Trajectory-Smoothness Curation Metrics Under Episode-Length Controls*

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Abstract

Trajectory smoothness has been proposed as a cheap, training-free signal for curating imitation-learning demonstrations, most recently by RINSE, which introduces Spectral Arc Length (SAL) and a contact-aware Trajectory-Envelope Distance (TED) and reports downstream gains on robomimic. Recent audit work showed that generic curation metrics on a synthetic-defect LIBERO testbed largely exploit episode length rather than demonstration content. We test whether the *specific published metrics* survive the corresponding controls on *real* data. On robomimic can/mh, SAL separates better from worse operators with AUROC 0.952, but episode length alone achieves 0.957, and inside the tightest duration-matched band SAL falls to 0.628 while raw mean jerk, the baseline RINSE argues against, rises to 0.896. On can/mg, where every episode is exactly 150 steps and length cannot carry information, SAL is at chance (0.454) and TED is inverted (0.376) against the task-success label, while jerk reaches 0.706. Training GMM behavior-cloning policies on each metric’s curated subset (10 seeds per condition), curating by TED costs roughly a third of downstream success relative to random selection of the same size (0.240 vs. 0.360, $p=0.0001$, surviving Bonferroni correction); path length is directionally harmful with weaker support, while SAL and jerk are not statistically distinguishable from not curating at all. Replicating the grid under a diffusion policy leaves TED ranked worst under both architectures, though at five seeds its deficit there is no longer resolvable, and reverses jerk from inert to significantly better than both baselines, so the curation decision does not transfer unchanged across the two policy configurations we tested. Inspecting individual trajectories shows why: on this task successful manipulation requires sustained high-deviation motion through the contact phase, while failure manifests as low-amplitude hovering that a smoothed envelope follows closely, so an envelope-based smoothness metric measures commitment with the sign flipped. The length confound replicates on a second task: on square/mh, SAL correlates +0.96 with episode duration and its skill AUROC falls from 0.898 to 0.480 inside a duration-matched band, while jerk rises from 0.431 to 0.896. Our results extend the detection-vs-curation decoupling to named, published metrics, on operator labels rather than injected defects, and to a benchmark where the length confound is excluded by construction rather than by truncation.

1 Introduction

Behavior cloning inherits the quality of its demonstrations, and a growing literature proposes to score and filter demonstrations automatically. Training-free trajectory statistics are the cheapest family of such scores, and the most developed recent proposal is RINSE [1], which ranks demonstrations by two smoothness metrics: Spectral Arc Length (SAL), a frequency-domain measure adapted from rehabilitation science, and Trajectory-Envelope Distance (TED), a contact-aware spatial deviation from a Bézier-smoothed reference. RINSE reports that filtering by these metrics matches or beats full-data training on robomimic tasks.

A recent line of audit work complicates this picture. Bedi [2] showed on a controlled testbed that action-only curation scorers are blind or inverted on structural defects, and the follow-up [3] showed that five of seven curation metrics on a LIBERO testbed achieved near-perfect detection AUROC by measuring episode length

*Code and scored data: <https://github.com/imjbassi/smoothness-metric-audit> (released on publication).

rather than demonstration content, and that detection accuracy and curation value are sharply decoupled. Those results, however, were obtained with *generic* metric implementations (a SPARC-style smoothness score among them), with *synthetically injected* defects, and with episode length controlled by *truncation*.

This paper asks the natural next question: do the specific, published metrics of RINSE survive these controls on real data? Our audit differs from the Bedi testbeds on three axes, each of which closes a gap that line of work explicitly left open:

1. **Named metrics, not proxies.** We implement SAL from its published formula and TED from the RINSE appendix, including the contact-boundary partition that is TED’s distinguishing design feature. TED has not previously been evaluated outside its authors’ own success-filtered setting.
2. **Real labels, not injected defects.** robomimic can/mh [6] carries *human operator skill tiers* (better/okay/worse) assigned to real teleoperators, and can/mg carries real task-success outcomes of a scripted-policy data-generation process. No defect is synthetic.
3. **A length-free benchmark by construction.** In can/mg every episode is exactly 150 steps. The episode-length confound documented by [3] cannot exist in this data, with no truncation required. Truncation removes the confound by discarding signal; a fixed-horizon dataset removes it without touching the trajectories.

The scope of the evidence is worth stating before the results rather than after them. The detection results, which involve no training, cover two robomimic tasks, Can and Square, and use real operator skill tiers as labels. The success-label results cover one task, Can, because it is the only robomimic task that ships splits containing failures. The downstream-training results also cover Can only, under two policy configurations: a GMM behavior-cloning policy, run for all four metrics at ten seeds, and a diffusion policy, run for two of the four metrics at five seeds. The downstream claims we make about TED are therefore established under behavior cloning only, and the claim we make about jerk under the diffusion policy holds in the configuration we tested. We make no claim about other tasks, about human teleoperation failures, or about other policy configurations, and Section 7 states what each extension would require.

Our findings are easy to state. Smoothness metrics rank *operator skill* largely to the extent that skill is confounded with how long the episode took. Once duration is controlled, by normalization or by duration-matched subsets, SAL’s advantage collapses and raw mean jerk, the crude baseline the smoothness literature positions itself against, becomes the strongest detector. Against *task success*, on fixed-length data, both of RINSE’s metrics are at or below chance, with TED inverted, while jerk retains signal. Curating by TED produces a behavior-cloning policy substantially worse than random subsampling of the same size, a result that survives correction for multiple comparisons; the weaker metrics are not distinguishable from doing nothing. Skill and success are different targets, and the curation literature validates on one while deploying against the other.

2 Related Work

Trajectory-statistic curation. RINSE [1] is the primary subject of this audit; it assumes demonstration pools pre-filtered for task success and validates on successful trajectories only, a scoping decision its limitations section states explicitly. PSD [5] scores demonstrations by total spectral power of the end-effector trajectory and reports gains over learning-based curation on robomimic-MH and a non-expert dataset. Neither work evaluates on pools containing failures.

Audits of curation metrics. Bedi’s testbed papers [2–4] established the detection-vs-curation decoupling, documented the episode-length confound, and reported a negative result on per-phase metric selection. We adopt their framing and extend it to published metrics, real labels, and a natively fixed-length benchmark. Where they control length by truncating to a fixed prefix, our can/mg experiments require no such intervention.

Evaluation practice. Kress-Gazit et al. [7] and RoboEval [8] argue that binary success rate under-describes policy quality, and [9] shows that widely used benchmarks admit shortcut solutions. Our results are a data-side complement: the scores used to *select training data* can also be shortcut-driven, with episode duration as the shortcut.

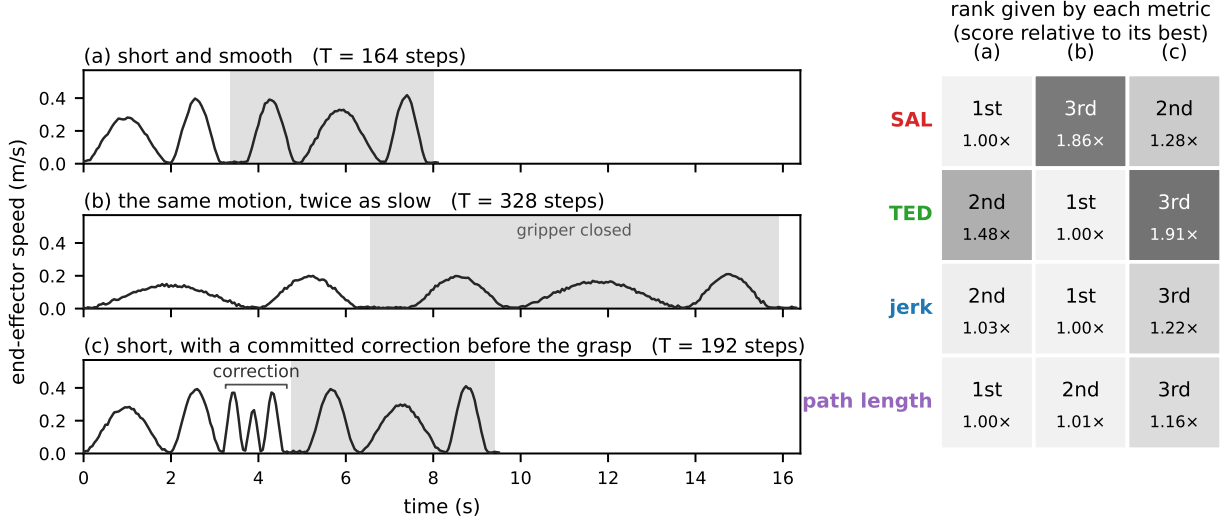


Figure 1: **What the four metrics reward, on illustrative synthetic data.** Left: end-effector speed profiles of three synthetic pick-and-place demonstrations built from minimum-jerk segments between identical waypoints, with the same small band-limited jitter added to each (shaded: gripper closed). (a) is short and smooth. (b) is the same motion executed twice as slowly, so the episode is twice as long and every segment is just as smooth. (c) is (a) with a committed correction inserted before the grasp: lift, shift sideways, descend again. Right: how each metric ranks the three, computed with the implementations used throughout the paper, with every score given relative to the best of the three under that metric. SAL rates the slow copy (b) worst, by a wider margin than it charges the correction in (c). TED rates the correction (c) worst and the slow copy (b) best. Jerk also rates (c) worst and barely separates (a) from (b). Path length cannot separate (a) from (b), which travel the same route. Every ordering shown is stable across 30 noise seeds except the (a)/(b) near-tie under jerk, which swaps in about a third of them, and the (a)/(c) order under SAL, which swapped once (`scripts/fig_metric_primer.py`). Synthetic and illustrative only: no number in this figure is measured on robomimic.

3 Metrics and Datasets

3.1 What each metric measures

Readers who already know these metrics can skip to Section 3.2. All four return a badness score, higher is worse, and curating by a metric means keeping the demonstrations with the lowest scores. The intuition behind each metric comes first here, and the formal definitions follow in Section 3.2.

SAL (Spectral Arc Length). SAL looks only at how fast the end effector is moving over time, ignoring where it goes. It takes that speed trace, breaks it into its frequency content, and measures how long and jagged the resulting spectrum is. A speed trace made of a few clean rises and falls, as in a single deliberate reach, has a spectrum that decays quickly and smoothly, which gives a short arc and a good score. A trace full of small accelerations and hesitations has content at many frequencies and a longer arc. The metric was developed in rehabilitation science to score the reaching movements of patients against those of healthy subjects, and the argument for applying it to demonstrations is that a skilled operator moves in a few committed strokes while an unskilled one makes many small adjustments.

TED (Trajectory-Envelope Distance). TED asks how far the recorded motion strays from a smoothed copy of itself. It cuts the episode into pieces at every moment the gripper opens or closes, fits each piece with a smooth curve (the envelope), and reports the average gap between the recorded pose and that curve along the best time alignment of the two. Motion that a smooth curve can follow closely scores well. Motion with excursions the curve cannot follow scores badly. The cuts at gripper events are what make TED contact-aware: the abrupt change of motion that necessarily accompanies a grasp or a release falls on

a segment boundary instead of being counted as a deviation, so what remains is meant to be the operator’s wobble within each phase of the task.

Jerk. Jerk is the rate at which acceleration changes. Mean jerk takes the third difference of the end-effector position at every timestep and averages its magnitude over the episode. Motion that speeds up and slows down gradually scores well. Abrupt starts, stops, and changes of direction score badly. It is the classical smoothness measure in the motor-control literature and the crude baseline that SAL and TED were proposed to improve on. The usual criticisms are that it amplifies sensor noise and that it does not distinguish a deliberate fast motion from a shaky one.

Path length. Path length is the total distance the end effector travels. A demonstration that goes more or less directly from start to goal scores well. One that wanders, backtracks, or makes repeated attempts scores badly. The intuition is that a skilled operator takes the direct route.

Which of these grow with episode length. Two of the four are tied to duration and two are not. SAL’s arc is a sum over the $T/2$ frequency bins of the spectrum, and on real data every bin carries some noise and adds to the sum, so recording the same motion for twice as long raises SAL even when nothing about the execution has changed. On can/mh and square/mh, SAL’s correlation with T is above +0.95 (Section 4). Path length grows with distance rather than time, so a slow retrace of the same route scores the same as a fast one, but an operator who takes longer usually also travels further, and its correlation with T is +0.86 on can/paired and +0.90 on square/mh. Jerk and TED are averages per timestep (TED per step of the time alignment) and do not accumulate with T . TED’s correlation with T on can/mh is +0.14. Slowing the same motion down lowers every acceleration along it, so jerk, if anything, favors slower execution of the same path.

Figure 1 shows the two behaviors that the rest of the paper turns on, using three synthetic speed profiles scored with our own implementations. Demonstration (b) is demonstration (a) executed twice as slowly, with every segment exactly as smooth, and SAL rates it the worst of the three, by a wider margin than it charges the committed correction in (c). SAL rewards shortness. TED rates the correction in (c) as the worst of the three and the slow copy in (b) as the best. TED penalizes committed motion. On real data, the first behavior appears as SAL tracking episode length almost exactly (Section 4), and the second as TED rating the low-deviation hovering of failed can/mg demonstrations as cleaner than the committed contact-phase motion of successful ones (Section 6).

3.2 Formal definitions, datasets, and statistics

Metrics. All metrics are computed from proprioceptive signals at the robomimic control rate of 20 Hz ($\Delta t = 0.05$ s) and return a badness score (higher is worse). *SAL*: spectral arc length of the end-effector speed profile, implemented from the published formula; we negate so that higher means less smooth. *TED*: contact-aware trajectory-envelope distance, reimplemented from the RINSE appendix: the 6-D pose trajectory (position plus rotation vector, orientation weight $w_{\text{ori}} = 0.25$) is partitioned at gripper open/close transitions, each segment is fit with a greedy Bézier envelope, and the score is the length-normalized DTW distance between trajectory and envelope. *Jerk*: mean third-difference magnitude of end-effector position. *Path length*: total end-effector arc length. Where indicated we also report per-timestep normalizations (score divided by $T-1$). Our TED is a reimplement from the paper, not the authors’ code; Section 7 discusses the resulting caveat and the sanity check we ran.

Datasets. All data is public robomimic [6] low-dim. *can/mh*: 300 successful demonstrations from six human operators with released skill tiers (better/okay/worse, 100 each); episode lengths range 98–1050 steps. *can/mg*: 3900 machine-generated demonstrations, 18.4% successful, every episode exactly 150 steps. *can/paired*: 200 demonstrations in matched success/failure pairs (50% successful), lengths 55–147; we report it as a secondary setting because its matched-pair construction makes it adversarially hard for trajectory-level scores. *square/mh*: 300 successful demonstrations with the same skill-tier structure (100 each), episode lengths 123–1051, used in Section 4.4 to test whether the length confound is task-specific. We note that Can is the only robomimic task with failure-containing splits (**mg** and **paired**); Square, Transport, and Tool Hang ship only proficient-human and multi-human data, all of it successful. Our success-label experiments are therefore confined to Can by benchmark availability, not by choice.

Statistics. All AUROCs are reported with 95% bootstrap confidence intervals (2000 resamples). Downstream comparisons use difference-of-means bootstrap CIs (20,000 resamples) and Welch two-sample tests,

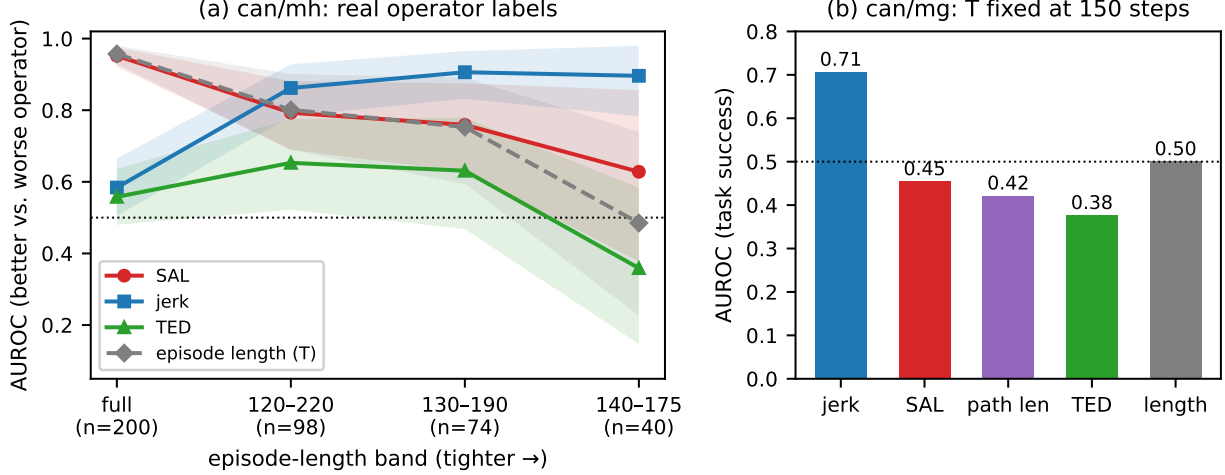


Figure 2: **Smoothness metrics rank duration, not quality.** (a) Better-vs-worse operator AUROC on can/mh across duration bands of increasing tightness (shaded: 95% bootstrap CI). SAL (red) tracks the episode-length baseline (gray, dashed) almost exactly and falls from 0.952 to 0.628 as the band tightens; raw jerk (blue) moves the opposite way, 0.584 to 0.896; TED (green) never clears chance and inverts in the tightest band. In that band episode length itself is at chance (0.485), confirming the control. (b) On can/mg every episode is exactly 150 steps, so length carries no information by construction; against the task-success label, jerk retains signal (0.706), SAL is at chance (0.454), and TED is inverted (0.376).

with Bonferroni correction applied across the full set of reported comparisons; we mark which results survive correction and which do not. For skill we report better-vs-worse AUROC; for success we report AUROC against the binary success label with scores negated so that $\text{AUROC} > 0.5$ means the metric ranks clean/successful demonstrations higher, and $\text{AUROC} < 0.5$ means the metric is *inverted*.

4 Detection Results

4.1 Operator skill on can/mh: the length confound, on real labels

On the full 200 better/worse demonstrations, SAL appears excellent: AUROC 0.952 [0.922, 0.975]. But episode length alone achieves 0.957 [0.929, 0.980], and SAL’s correlation with T is +0.97. The tier means make the mechanism plain: worse operators average 304 steps, better operators 143, and SAL’s tier means (100.4 / 57.5 / 44.9) track duration almost linearly. On this dataset SAL is, to first order, a stopwatch.

Two independent controls confirm this. First, *normalization*: SAL’s arc length sums over $T/2$ spectral bins, so dividing by $T-1$ is the principled per-timestep correction; normalized SAL drops from 0.952 to 0.582 [0.497, 0.662] and its length correlation drops from +0.97 to +0.07. Second, *duration matching*: restricting to bands of increasingly similar episode length (Figure 2a, Table 1). In the tightest band ($T \in [140, 175]$, $n=40$), where length itself is at chance (0.485 [0.227, 0.740]), SAL is 0.628 [0.378, 0.856], a CI that includes chance, while raw jerk is 0.896 [0.783, 0.980]. Across the four bands the two metrics *cross*: as the confound is removed, the crude metric strengthens monotonically (0.584, 0.862, 0.906, 0.896) and the sophisticated one weakens monotonically (0.952, 0.793, 0.759, 0.628).

TED never clears chance in any band (0.558 full-sample, upper CI 0.636) and inverts to 0.359 in the tightest band. TED does *not* have the length pathology (correlation +0.14 with T); it simply carries little skill signal on this data.

We caution that the tightest band retains only 7 worse-operator demonstrations and its intervals are wide; the claim rests on the *trend* across all four bands and its agreement with the normalization control, not on any single cell. We also note that naïve per-timestep normalization is only appropriate for metrics whose raw value grows with T ; normalized path length, for example, becomes an inverse-speed proxy that

	full ($n=200$)	$T \in [120, 220]$ ($n=98$)	$T \in [130, 190]$ ($n=74$)	$T \in [140, 175]$ ($n=40$)
SAL	0.952 [.922,.975]	0.793 [.690,.881]	0.759 [.628,.875]	0.628 [.378,.856]
SAL/ T	0.582 [.497,.662]	0.600 [.479,.715]	0.603 [.446,.755]	0.658 [.342,.923]
TED	0.558 [.480,.636]	0.653 [.522,.776]	0.631 [.468,.779]	0.359 [.147,.583]
jerk	0.584 [.505,.665]	0.862 [.783,.928]	0.906 [.832,.965]	0.896 [.783,.980]
path len.	0.783 [.718,.842]	0.590 [.449,.730]	0.624 [.461,.781]	0.429 [.215,.645]
length T	0.957 [.929,.980]	0.801 [.689,.901]	0.753 [.594,.891]	0.485 [.227,.740]

Table 1: Better-vs-worse operator AUROC on can/mh across duration bands, with 95% bootstrap CIs. The bottom row is the control: a band is duration-matched only where T itself is near chance. Per-timestep normalizations of TED and path length are omitted: unlike SAL, whose raw value grows with the number of spectral bins, these quantities do not scale linearly with T , and dividing by T turns them into inverse-speed proxies rather than duration-free measures of smoothness.

rises with skill and is reported only for completeness in Table 1.

4.2 Task success on can/mg: the confound removed by construction

The can/mg split fixes every episode at 150 steps, so no metric can exploit duration. Against the success label (716 of 3900 successful), jerk achieves 0.706, SAL 0.454, path length 0.421, and TED 0.376 (Figure 2b). The length baseline is exactly 0.500, as it must be. Per-timestep normalization leaves every AUROC unchanged, as expected under a constant divisor, which we report as a consistency check.

Two readings of this result matter. First, both of RINSE’s metrics are at or below chance on the label practitioners actually deploy against. TED’s inversion means that filtering by TED *preferentially retains failed demonstrations*. Second, the ordering among metrics reverses relative to the uncontrolled skill setting: the metric that looked worst there (jerk) is the only one with signal here.

4.3 Secondary setting: can/paired

On the matched-pair dataset every metric is inverted against success (TED 0.331, SAL 0.398, jerk 0.368, path length 0.029), but the setting is confounded in the opposite direction: successful demonstrations are systematically *longer* (length-alone AUROC 0.194), and path length’s +0.86 correlation with T makes it a pure length detector here. We report can/paired for completeness and treat can/mg as the primary success-label evidence.

4.4 Generalization to a second task: square/mh

If the length confound were an artifact of the Can task, the argument would be considerably weaker. We therefore repeat the skill-label analysis on square/mh, a harder insertion task with the same operator skill-tier structure and a comparable duration range (123–1051 steps versus Can’s 98–1050). No retuning was performed: the same metric implementations and the same band procedure were applied to the new data.

The confound replicates almost exactly. SAL’s correlation with episode duration is +0.96 on Square against +0.97 on Can, and its uncontrolled skill AUROC is 0.898 [0.852, 0.936] against a length-alone baseline of 0.936 [0.902, 0.964]. Per-timestep normalization collapses it to 0.458 [0.376, 0.536], slightly below chance and a larger drop than the corresponding Can figure. Path length behaves the same way, reaching 0.931 [0.893, 0.963] uncontrolled with a +0.90 duration correlation.

The crossover also replicates (Table 2). As the duration band tightens and the length baseline falls from 0.936 to 0.407, SAL falls monotonically (0.898, 0.759, 0.529, 0.480) while jerk rises monotonically (0.431, 0.570, 0.782, 0.896). In the band where duration is genuinely controlled ($T \in [200, 240]$, length AUROC 0.407), jerk reaches 0.896 [0.724, 1.000], numerically identical to its value in Can’s tightest band, while SAL sits at 0.480 [0.250, 0.701], a CI centred on chance.

Two differences from Can are worth stating. First, jerk is *inverted* on Square before any control is applied (0.431), where on Can it was weakly positive (0.584); the uncontrolled ranking of metrics is therefore

	full ($n=200$)	$T \in [150, 300]$ ($n=111$)	$T \in [180, 260]$ ($n=55$)	$T \in [200, 240]$ ($n=30$)
SAL	0.898 [.852,.936]	0.759 [.658,.849]	0.529 [.344,.702]	0.480 [.250,.701]
SAL/ T	0.458 [.376,.536]	0.485 [.372,.602]	0.468 [.300,.653]	0.520 [.280,.738]
TED	0.663 [.584,.736]	0.685 [.577,.788]	0.599 [.427,.758]	0.557 [.335,.785]
jerk	0.431 [.349,.509]	0.570 [.455,.687]	0.782 [.618,.923]	0.896 [.724,1.000]
path len.	0.931 [.893,.963]	0.817 [.726,.895]	0.796 [.654,.913]	0.887 [.742,.991]
length T	0.936 [.902,.964]	0.824 [.738,.898]	0.613 [.456,.766]	0.407 [.197,.629]

Table 2: Better-vs-worse operator AUROC on *square*/mh across duration bands, with 95% bootstrap CIs, in the same format as Table 1. The bottom row is the control. The SAL/jerk crossover observed on Can reproduces here: SAL tracks the length baseline down while jerk rises, and in the band where length is at chance jerk reaches the same 0.896 it reaches in Can’s tightest band.

condition	pool size	best success rate (mean $\pm$ std)	95% CI
oracle (successes only)	716	0.892 $\pm$ 0.048	[0.866, 0.922]
no curation	1950	0.380 $\pm$ 0.063	[0.344, 0.420]
random 50%	1950	0.360 $\pm$ 0.061	[0.326, 0.398]
SAL	1950	0.332 $\pm$ 0.061	[0.298, 0.370]
jerk	1950	0.332 $\pm$ 0.041	[0.306, 0.354]
path length	1950	0.310 $\pm$ 0.030	[0.292, 0.328]
TED	1950	0.240 $\pm$ 0.045	[0.212, 0.266]

Table 3: Downstream GMM-BC policy success on can/mg after filtering to a fixed 1950-demonstration pool by each metric, 10 seeds each. The oracle pool is smaller (716, the number of successes that exist) and is a ceiling reference, not a matched condition. Pairwise tests against the baselines are in Table 4.

itself task-dependent, even though the controlled ranking is not. Second, path length remains strong in the controlled band on Square (0.887 [0.742, 0.991]) where it collapsed on Can (0.429 [0.215, 0.645]). We do not claim jerk or path length is a good curation metric on the strength of these numbers; the point is narrower, that whatever advantage SAL displays over them disappears once duration is held fixed, on both tasks. The tightest Square band retains 13 worse-operator demonstrations, so as with Can the claim rests on the monotone trend across bands rather than on any single cell.

5 Does Detection Predict Curation? Downstream Training

Detection AUROC is not the quantity practitioners care about [3]; the trained policy is. We therefore filter can/mg by each metric at a 50% keep fraction and train a GMM behavior-cloning policy (MLP 1024×1024 , 5 modes, batch 100, 600 epochs of 500 gradient steps, learning rate 10^{-4}) on each curated subset, with 50 evaluation rollouts (horizon 400) every 150 epochs and the best checkpoint reported per seed. Conditions: TED, SAL, jerk, path length, plus three references: an *oracle* keeping only successful demonstrations, a *random* 50% subset, and an uncured size-matched control. Ten seeds per condition. All filtered pools contain 1950 demonstrations except the oracle (716, the number of successes that exist); we therefore treat the oracle as a ceiling reference rather than a matched condition.

Table 3 answers the question Section 4 could only pose, and Table 4 reports the comparisons that matter. Two results are unambiguous. Curating by TED costs roughly a third of downstream success relative to random selection of the same size (-0.120 , $p = 0.0001$) and relative to not curating at all (-0.140 , $p < 0.0001$); both survive Bonferroni correction across the eight comparisons we report. Path length is directionally harmful with weaker support ($p = 0.037$ and $p = 0.0071$), clearing conventional significance but not correction. SAL and jerk sit below both baselines in the mean but are not statistically distinguishable from them ($p \geq 0.06$), and we report them as such rather than as evidence of harm.

We emphasize this because an earlier version of this analysis, run with 5 seeds rather than 10, appeared to

metric	vs. random 50%		vs. no curation	
	Δ [95% CI]	p	Δ [95% CI]	p
TED	-0.120 [-0.166, -0.076]	0.0001	-0.140 [-0.186, -0.096]	< 0.0001
path length	-0.050 [-0.090, -0.012]	0.037	-0.070 [-0.112, -0.030]	0.0071
jerk	-0.028 [-0.072, +0.014]	0.25	-0.048 [-0.092, -0.006]	0.060
SAL	-0.028 [-0.080, +0.024]	0.32	-0.048 [-0.098, +0.002]	0.098

Table 4: Difference in mean best success rate between each metric-curated condition and the two baselines, with 95% bootstrap CIs on the difference (20,000 resamples) and Welch two-sample p -values. Bold marks the comparisons surviving Bonferroni correction across all eight tests ($\alpha = 0.05/8 = 0.006$). Only TED clears that bar. For reference, random and no-curation are themselves indistinguishable ($\Delta = -0.020$, CI $[-0.072, +0.032]$).

show every metric-curated condition underperforming both baselines. That pattern did not survive the seed expansion for SAL and jerk. Since the argument of this paper is that under-powered evaluation produces confident claims in the wrong direction, we apply the same standard to our own results: the defensible claim is about TED, not about smoothness metrics as a class.

What the grid does establish irrespective of any single comparison is that under this policy configuration none of these metrics recovers usable signal that is demonstrably present. An oracle keeping only true successes reaches 0.892 [0.866, 0.922], roughly $2.5\times$ the best metric-curated condition, so the pool contains ample information about which demonstrations to keep; the trajectory statistics simply do not find it. Random subsampling and the uncured control are themselves indistinguishable ($\Delta = -0.020$, CI spanning zero), which is the sanity check one wants: at a 50% keep fraction on this data, discarding demonstrations at random costs nothing measurable, so any deficit a metric shows is attributable to *which* demonstrations it discards rather than to the reduction in pool size.

The downstream ordering tracks the detection ordering imperfectly. TED is worst in both, and path length is weak in both, but jerk has the best detection AUROC on can/mg (0.706) while landing indistinguishable from SAL downstream, whose detection AUROC is near chance (0.454). Detection accuracy is at best a coarse predictor of curation value, consistent with the decoupling reported by [3] on a different testbed.

5.1 Cross-architecture replication under a diffusion policy

The grid above uses a single policy configuration, so on its own it cannot distinguish a property of the metric from a property of behavior cloning. We therefore replicated it under a diffusion policy on the same curated subsets: same keep fraction, same 1950-demonstration pools, same 600 epochs of 500 gradient steps, same learning rate (10^{-4}), and the same evaluation protocol (50 rollouts, horizon 400, every 150 epochs, best checkpoint per seed).¹ The policy is a conditional U-Net ([256, 512, 1024] down-dims, kernel 5, 8 groups, 256-dimensional diffusion-step embedding) with EMA (power 0.75), an observation horizon of 2, action horizon of 8, and prediction horizon of 16, using a DDIM noise scheduler with 10 inference steps. Batch size (256 vs. 100) and the sequence modeling are inherent to the architecture and are the only respects in which the two grids differ by more than the policy configuration. We ran five conditions (TED, jerk, and the three references) with five seeds each; SAL and path length were not rerun, so the diffusion comparison covers two of the four metrics.

Three results follow, and they do not all point the same way.

First, TED is the worst condition under both architectures (0.240 under BC, 0.092 under diffusion). That is what one expects if the deficit belongs to the metric rather than to behavior cloning, and the replication is

¹Reporting the best checkpoint per seed is a maximum over the eight evaluations each run receives, and is therefore biased upward; both grids are evaluated eight times, so the bias is symmetric across architectures. Recomputing every comparison with the *mean* over the eight evaluations instead leaves all conclusions intact, with one exception we record here: TED versus no curation under diffusion reaches $p = 0.0110$ under the mean estimator and would survive this architecture’s correction threshold ($\alpha = 0.0125$), whereas under the reported best-of-eight estimator it does not ($p = 0.11$). TED versus random remains non-significant under both ($p = 0.12$ and $p = 0.17$). We report the conservative version throughout.

condition	GMM-BC (10 seeds)	diffusion (5 seeds)
oracle (successes only)	0.892 $\pm$ 0.048	0.544 $\pm$ 0.030
no curation	0.380 $\pm$ 0.063	0.120 $\pm$ 0.014
random 50%	0.360 $\pm$ 0.061	0.116 $\pm$ 0.017
jerk	0.332 $\pm$ 0.041	0.212 $\pm$ 0.030
TED	0.240 $\pm$ 0.045	0.092 $\pm$ 0.030

Table 5: Mean best success rate under both policy configurations, restricted to the five conditions run under both. TED is the worst condition under each. Jerk moves from indistinguishable-from-baseline under BC to above both baselines under diffusion. Absolute rates are lower under diffusion in every condition, by factors ranging from $1.6\times$ (oracle) to $3.1\times$ (the two baselines).

metric	vs. random 50%		vs. no curation	
	Δ [95% CI]	p	Δ [95% CI]	p
jerk	+0.096 [+0.068, +0.124]	0.0007	+0.092 [+0.064, +0.116]	0.0011
TED	-0.024 [-0.048, +0.004]	0.17	-0.028 [-0.052, +0.000]	0.11

Table 6: Diffusion-policy grid, same procedure as Table 4: 95% bootstrap CIs on the difference (20,000 resamples) and Welch two-sample p -values. Bold marks comparisons surviving Bonferroni correction across the four tests reported for this architecture ($\alpha = 0.05/4 = 0.0125$). Random and no-curation are again indistinguishable ($\Delta = -0.004$, CI $[-0.020, +0.012]$).

consistent with that part of the original finding. But the harm is statistically robust under BC only. Under diffusion the deficit is -0.024 against random and -0.028 against no curation, with intervals that admit zero, and neither survives correction. This is a statement about power rather than a refutation: at five seeds this grid resolves differences of roughly 0.054, essentially the same threshold we report for the BC grid, and TED’s deficit falls below it. The defensible claim is that TED’s *ordering* holds across architectures while its *harm* is demonstrated under BC only. Descriptively, and not as a tested comparison, the deficit is of broadly similar relative size in the two grids: 37% of the no-curation baseline under BC and 23% under diffusion. We report this to characterize how the compressed diffusion success scale relates to the absolute differences in Table 6, not as evidence that the two effects are equal.

Second, jerk reverses sign. Under BC it is indistinguishable from both baselines ($p \geq 0.06$); under diffusion it beats both by $+0.096$ and $+0.092$, and both comparisons survive correction. This is the largest effect in the diffusion grid, and it runs opposite to the paper’s headline: at least one crude trajectory statistic, never proposed as a curation metric, has genuine curation value in the configuration we tested. It is also the second time in this work that jerk outperforms the metrics motivated against it, the first being detection inside a duration-matched band. What may drive the reversal is taken up below.

Third, absolute success rates are lower under diffusion in every condition, by factors ranging from $1.6\times$ on the oracle (0.544 vs. 0.892) to $3.1\times$ on the two baselines, at a matched epoch and evaluation budget in which the diffusion runs in fact saw more samples per epoch (batch 256 vs. 100). The gap is therefore not a uniform offset: it is widest exactly on the mixed-quality pools the curation comparison runs on.

One explanation can be ruled out. The diffusion runs show no measurable improvement over the second half of training: averaged across all 25 runs, mean success in the last four evaluations exceeds the first four by $+0.017$, against a binomial noise floor of 0.044 for 50-rollout evaluations, and only 3 of 25 runs improve by more than one noise standard deviation. Per-run variation across the eight evaluations has a standard deviation $1.01\times$ the noise floor, which is to say it is indistinguishable from sampling noise alone. The identical measurement on the BC grid gives $+0.092$, with 56 of 70 runs improving and a ratio of $1.46\times$, so the flatness is a property of the diffusion runs and not of our evaluation protocol. Whatever limits these policies is therefore not training duration; they converge early and stop. We record this because under-training is the most natural explanation for the gap and it does not survive contact with the training curves, which narrows the space of remaining explanations without confirming one. One candidate is action parameterization: our

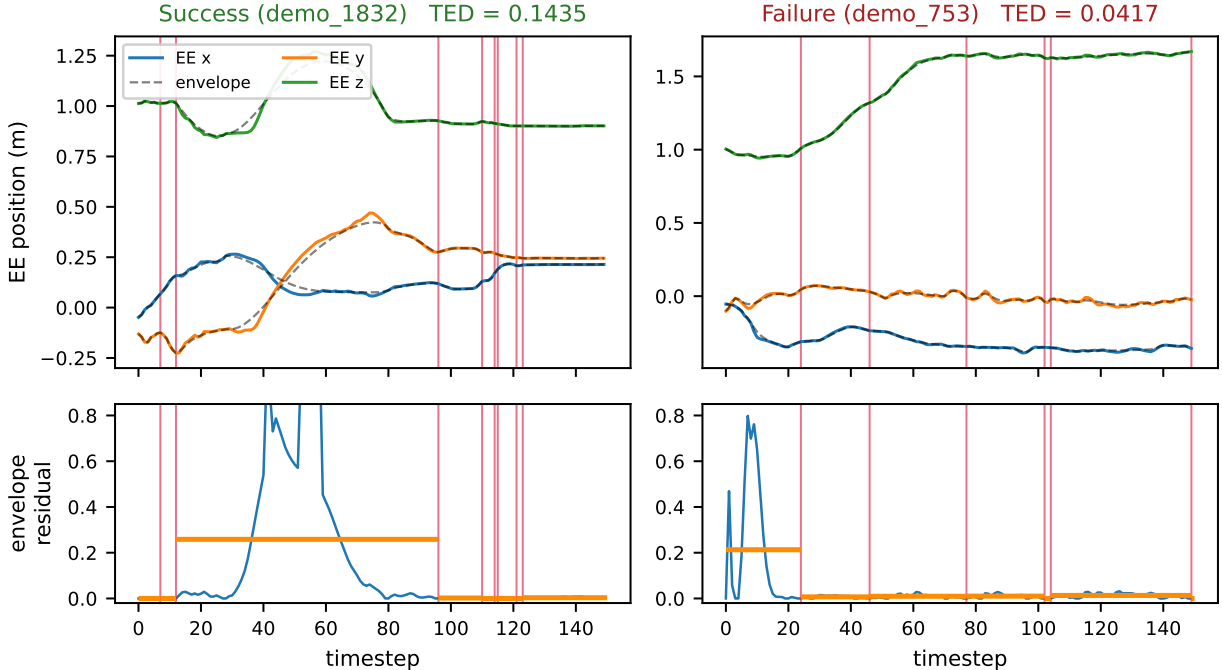


Figure 3: **Failure stays inside the envelope; success does not.** A successful (left) and failed (right) can/mg demonstration, with end-effector position, TED’s gripper-contact segment boundaries (red), the per-segment Bézier envelope (dashed), and the per-timestep envelope residual (bottom, orange bars are per-segment means). Both trajectories contain one high-deviation segment, but in the success it spans timesteps 12–96 and coincides with the descend-and-grasp, while in the failure it is confined to the initial free-space approach (0–24) and the remaining 126 steps are a flat tail of repeated gripper cycles that never completes a placement. TED averages over the alignment path and so rates the failure as the cleaner trajectory. Illustrative of the mechanism only; the quantitative claim is Figure 2b.

runs used robomimic’s default delta actions, which Chi et al. [10] report handicap diffusion policy while helping BC, so the gap and the jerk reversal may partly reflect it, and the policy-class reading of both is provisional. The consequence for reading Table 5 is concrete: the two columns are not operating at comparable points on the success scale, so cross-architecture comparisons of absolute differences are weaker than within-column comparisons, and the claims above are deliberately about ordering and sign rather than magnitude.

We correct within each architecture, since each grid is a separate family of comparisons. Pooling all twelve tests instead ($\alpha = 0.05/12 = 0.00417$) changes nothing: TED’s two BC comparisons and jerk’s two diffusion comparisons still survive, and path length’s $p = 0.0071$ still does not.

Taken together, the replication is consistent with the metric-level reading of the TED result and simultaneously narrows the scope of any claim about smoothness metrics as a class. Which demonstrations are worth keeping is not a property of the demonstrations alone, and a curation result reported without naming the policy configuration it was measured on, architecture and action parameterization included, omits something that can change the sign of the answer.

6 Discussion

Skill and success are different targets. The same metric family gives opposite verdicts depending on the label. Under an uncontrolled skill label, SAL looks nearly perfect; under a length-controlled skill label it is marginal; under a success label on fixed-length data it is at chance and TED is inverted. Curation papers validate against operator skill or smoothness-adjacent labels on success-only pools, then are deployed, in

practice, to remove *failures* from mixed pools. Our results say those are not the same job, and the metrics do not transfer between them.

The crude baseline wins under controls. RINSE motivates SAL and TED as principled improvements over naïve statistics such as jerk. Under every length control we applied, that ordering reverses. We do not claim jerk is a good curation metric in general, and the Bedi testbeds [2] show smoothness-family scorers failing badly on structural defects; we claim only that on real robomimic data the published sophistication does not survive the controls that the audit literature has shown to be necessary.

Why TED inverts: failure stays inside the envelope. TED’s design premise is that partitioning at gripper-contact boundaries isolates task-relevant abruptness, and a plausible hypothesis going in was that this would let TED catch failures, since canonical structural failures are contact-boundary events. The data supports a different and simpler mechanism. Figure 3 contrasts a successful and a failed can/mg demonstration. Both contain one high-deviation segment, but they differ in when it occurs and how much of the episode it spans. In the success, the high-residual segment runs from timestep 12 to 96 (84 of 150 steps, mean residual 0.259) and coincides with the descend-and-grasp: the end effector dips, the lateral axes swing, and the envelope cannot track the motion. In the failure, the only comparable deviation is the initial free-space approach (timesteps 0–24, mean residual 0.213); after that the trajectory settles into a flat tail in which the end effector holds height while the gripper opens and closes repeatedly without completing a placement, and every remaining segment has residual below 0.014. Because TED averages deviation over the DTW alignment path, a committed motion spanning half the episode raises the score substantially while a short approach transient is diluted by the smooth remainder. TED therefore rates the failure as the cleaner trajectory (0.042 vs. 0.144).

On this task, in other words, successful manipulation *requires* sustained high-deviation motion through the contact phase, while failure manifests as low-amplitude hovering that the envelope follows closely. A metric that rewards agreement with a smoothed envelope is measuring commitment with the sign flipped. Jerk sees the same two behaviors from the other side, which is why it is the one metric that retains signal in Section 4. A committed motion is large but smooth: it carries the end effector far from the envelope, but its accelerations change gradually, so its envelope deviation is high and its jerk is low. Hovering is small but jittery: the end effector holds position through repeated gripper cycles, and what little it moves, it moves abruptly, so its envelope deviation is low and its jerk is high. The two demonstrations in Figure 3 show both readings at once, with the success at a mean jerk of 71.6 against 124.1 for the failure while TED rates the failure cleaner, and across the whole split successful demonstrations have lower mean jerk than failed ones (85.7 vs. 112.8), which is the direction the 0.706 AUROC reports. We stress that these panels illustrate a mechanism rather than establish it; the quantitative claims are the AUROCs of 0.376 and 0.706 in Figure 2b, and the per-demonstration TED scores we inspected (successes 0.144 and 0.079, failures 0.042, 0.042, and 0.017) are consistent with but not evidence for those numbers.

7 Limitations

Our detection results cover two robomimic tasks (Can and Square) but our success-label and downstream-training results cover only Can, because Can is the only robomimic task shipping failure-containing splits; extending the success-label analysis requires datasets with real, labeled teleoperation failures, such as REASSEMBLE [12] or DROID [11]. Our downstream results cover two policy configurations (GMM behavior cloning and a diffusion policy); the detection results are policy-free. The cross-architecture grid of Section 5.1 carries three limitations of its own. It covers two of the four metrics, since SAL and path length were not rerun under diffusion, so the claim that TED ranks worst under both architectures rests on a five-condition subset on the diffusion side against a seven-condition grid on the BC side. It uses five seeds rather than ten, resolving differences of roughly 0.054; TED’s diffusion deficit (-0.028 , CI $[-0.052, +0.000]$) falls below that, which is why we report its ordering as replicating and its harm as established under BC only. And absolute success rates remain lower under diffusion in every condition at a matched epoch and evaluation budget, from $1.6\times$ on the oracle (0.544 vs. 0.892) to $3.1\times$ on the two baselines. We previously attributed that gap to under-convergence in a reduced-compute pilot; restoring the full epoch and rollout budget did not close it, and the training curves rule that explanation out directly, since the diffusion runs improve by only $+0.017$ between the first and second halves of training against a 0.044 noise floor while the

BC runs improve by +0.092. The gap is therefore not a matter of training duration; action parameterization (Section 5.1) is a candidate we have not yet tested. Because the two grids sit at different points on the success scale, we restrict cross-architecture claims to ordering and sign rather than magnitude. Multiple-comparison correction is applied within each architecture, as each grid is a separate family; pooling all twelve tests ($\alpha = 0.00417$) leaves every reported conclusion unchanged. Our TED is a reimplement from the RINSE appendix, not the authors’ released code; as a sanity check it broadly agrees with SAL’s ranking on the clean proficient-human data, but a discrepancy with the reference implementation cannot be excluded, and we will replace it if code becomes available. The tightest duration band retains only 7 negatives, and its role in our argument is corroborative, not primary. The BC downstream conditions use 10 seeds; at this power, differences smaller than roughly 0.05 in mean success rate are not resolvable, which is why we report SAL and jerk as indistinguishable from the baselines rather than as harmful under that architecture. The oracle training condition has a smaller pool (716 vs. 1950) and is reported as a ceiling, not a matched comparison. Finally, our success-label experiments use machine-generated failures (can/mg); failure modes of human teleoperators may present differently. We report can/paired only as a secondary setting because its matched-pair construction, in which each success/failure pair shares a near-identical trajectory prefix and diverges only in the final placement, concentrates the discriminating signal into a handful of terminal timesteps and simultaneously introduces an opposing length confound (successful demonstrations are systematically longer, length-alone AUROC 0.194); neither property is representative of the mixed-quality pools curation metrics are deployed on.

8 Conclusion

Published trajectory-smoothness curation metrics, evaluated exactly where their authors did not, on real operator labels under duration controls and on fixed-length data against task success, rank episode duration more than demonstration quality. Once duration is controlled, the crude jerk baseline they were designed to supersede outperforms them, and the contact-aware metric inverts, and the same crossover reproduces on a second task without retuning. Training on the metrics’ own curated subsets shows the practical stakes are real but narrower than a five-seed pass suggested: curating by TED costs roughly a third of downstream success against random selection of the same size, a result robust to correction for multiple comparisons, while the other metrics are directionally below the baselines without being distinguishable from them. An oracle reaches 0.892 on the same pool, so the signal these metrics fail to find is unambiguously present in the data. In the diffusion configuration we tested, TED still ranks worst under both architectures, consistent with a metric-level rather than architecture-level reading of that result, though at five seeds its deficit there is not resolvable and the harm remains established under behavior cloning only. The same replication reverses jerk from inert to better than both baselines. A curation metric’s value, and its sign, depend on the policy configuration it is measured against, which is not currently part of how such results are reported.

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